



Introduction to Linear Regression Transcript

Hello everyone, welcome to lesson 1 video 1. In this lesson, we will introduce simple linear regression. Regression helps us to describe and quantify the relationship between two variables, one dependent and the other one independent. Have you ever looked at a scatter plot of data points and thought, it looks like there is a pattern there.

Today we are going to talk about one of the most important and most common ways to find pattern. This section introduces us to linear regression. Linear regression is linear association of variables.

Now when we call it as simple linear regression, it becomes linear association of one variable with one other variable. So we have one output variable and one input variable, but we are measuring relationship of two variables. One of them is output, one of them is input.

So that becomes a simple linear regression. And I will tell you all of these things in much details in later sections. So let's get started.

Generally in linear regression, the goal is either to make a prediction by using the equation that we have learnt from the data or we understand the relationship between variables. Let's say if you would want to understand what is the relationship between ice cream consumption and the rising levels of temperature in the city. So you might want to understand the relationship or you might want to understand how does the consumption happens when tomorrow the levels are going to increase and hence prediction becomes important.

But let's say in some of the cases, we would want only to understand how does the relationship looks like, how does one variable is related to another variable. So that's when we would want to understand things. And this example, which is shown here, you have university GPA as y axis, you have high school GPA as your x axis.

And all of these data points are nothing but GPA and both the GPAs. So let's say for example, this data point, it is 3.2 as high school GPA and 3.5 as university GPA. Now, you see that out of these scores that we have got, you see a simple trend and this trend looks linear as well.

You can easily fit a line on this trend. That's why it is called as linear regression. Now, you have one output variable or let's say one variable university GPA, which you are studying as association with high school GPA.

So how does university GPA is related to high school GPA. Now, there could be two different things that you can do with that relationship. One, use that relationship to predict.

Once you have got, let's say, university GPA is 1.2 times high school GPA, then you can use that relationship to predict anyone's university GPA given you already know their high school GPA. The second thing that you can also do is to understand what is

the relationship, to understand how high school and university GPA are related to each other. Now, when we talk about simple linear regression, and regression is all about association between variables, we have two different types of variables.

One dependent variable, the other is independent variable. Dependent variable is the variable that you want to predict or explain. This is also called as response or outcome variable.

Independent variable, on the other hand, is using what information or what variable do you want to make the predictions or explain the why variable. You need something to explain. Those explanatory variables, those input variables, those predictor variables are called as independent variables.

Then, what we do is, we try to make a simple relationship or assume a simple relationship like y is equal to fx where the function f in this case is thought to be or we underlay this assumption that it is linear because you will later on see that we assume a linear equation. Now, this is the equation of line. That's what I was talking to you about, linear equation.

You have y is equal to β_0 plus $\beta_1 x$. We get the data and data might look like the data which is here. On this data, we fit in y is equal to β_0 plus $\beta_1 x$.

Now, what is β_0 and what is β_1 ? β_0 is the predicted value of y when x is 0 while β_1 is the most important part. It tells us how much y is expected to change when x changes by one unit. So what is the rate of change of y with respect to x ? That's also what you can think of.

When you take derivative of y with respect to x , it becomes β_1 which is nothing but how does y changes with respect to x . Now, this is a simple place to visualise how does changing intercept and slope changes the equation of line. If I vary the intercept term which is β_0 , the idea is to fit a line which is closest to all of these data points. This is the same data that I took in the university GPA and your high school GPA.

Now if I use intercept as equals to 0, I might be far away from the entire cloud of data points. So my idea is to fit in a line, this might be the closest possible approximate of the line which is nearest to all the data points. Now once I fix the intercept, I can also vary the slope.

So I can vary both of them if you think of and then I can get to the best possible line. So these two are called as the parameters of the model which we generally try to get to get the relationship between y and x . So these are the two parameters right now if you see on the right hand side, the regression equation looks like 0.8 plus $0.8x$. Now this could be the best possible equation that we can fit in this data, best possible linear equation that we can fit in this data. The relationship may very well could be non-linear as well.

But linear regression helps us understand what is the linear or linear association between the two variables or the data point of two variables that we have. If we choose



a line which is far away, you might get a line equation like this. But then you will see that the error that you get because this does not explain the relationship between y and x , the error that you will be getting because of this line using this line for prediction.

Let's say for example a new value comes in at 4.5, you will be making a prediction which is here. But the prediction may very well following all of this data may very well lie somewhere here right. So which would be far away from what the actual value is.

So your predictions are going to be very very inappropriate or wrong if you follow a wrong equation. So the idea is to fit in a line in the data which is as close to all the data points as possible. Now the full model that we assume is a part of two things, one what we can explain using the data and what we cannot explain which is also called as the error term.

So this particular part that we can explain with the data is called as explainable part and this is what our prediction is all about right. So we use up to this part to make the prediction but our data generally if you think of data points generally they are confounded as in they also contains the error in it. They do not have the real values.

There could be noise terms as well and because of which some of those variations you can explain but you may not be able to explain some other variations as well right but this is the part of the variation of y which doesn't get explained which is the error term or noise. Now the idea is to find the line which best fits the cloud of data points okay which is as close to all the data points as possible. In summary linear regression helps us in various different ways to quantify one variable with another.

In the next video we will see how to use least square methods to estimate the regression line.

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